

DELIVERABLE 1 – SUBMISSION CHECKLIST

High Level Design

If you cannot tick a box, you are not yet ready to submit.

☒ A. Design Options (Exploring the Solution Space)

- ☐ I present multiple plausible engineering solution options, not only one idea.
 - ☐ Each option is described clearly enough that an engineer could understand how it would work at a conceptual level.
 - ☐ Each option addresses the user requirements defined in D1.
 - ☐ Each option is technically feasible within the discipline (not speculative or unrealistic).
 - ☐ The options represent genuinely different approaches, not minor variations of the same idea.
 - ☐ The options demonstrate engineering thinking, not product shopping or technology listing.
- ☐ If your options are simply *different brands, models, or components* → this is not engineering design.

☒ B. Engineering Trade-Off Analysis

- ☐ I define clear engineering criteria used to evaluate the design options. Typical criteria may include:
 - Performance
 - Power consumption
 - Reliability
 - Cost
 - Complexity
 - Maintainability
 - Scalability
 - Environmental impact
 - ☐ The comparison between options is structured and justified, not based on opinion.
 - ☐ Trade-offs are explicitly discussed (e.g., cost vs performance, complexity vs reliability).
 - ☐ The evaluation process is transparent and reproducible.
 - ☐ I explain why certain options are rejected.
 - ☐ I justify why the selected option is preferred.
- ☐ If the selected solution appears without showing how alternatives were evaluated → the design is not defensible.

☒ C. Selected High-Level System Design

- ☐ I clearly identify the selected design approach.
 - ☐ I explain why this design best satisfies the user requirements.
 - ☐ I provide a high-level architecture of the system.
 - ☐ The system is represented using engineering diagrams, such as:
 - Functional block diagram
 - System architecture diagram
 - Data flow diagram
 - User interfaces
 - ☐ The architecture describes system functions, not components.
 - ☐ The design demonstrates systems thinking, not isolated parts.
- ☐ If your architecture jumps directly to components, circuits, or algorithms → you have moved too far ahead (D3).

☒ D. Subsystem Identification

- ☐ I have identified the major subsystems required in the system.
- ☐ Each subsystem has a clear engineering purpose.
- ☐ Each subsystem description explains:
 - its function
 - its inputs

- its outputs
- its role in the overall system

☐ Subsystems represent functional units, not individual parts. Example subsystems may include:

- sensing subsystem
- control subsystem
- communication subsystem
- power subsystem
- user interface subsystem

☐ Listing components (e.g., “microcontroller”, “sensor”, “battery”) instead of subsystems → incorrect level of abstraction.

☒ E. Subsystem Interfaces

☐ I describe how subsystems interact with each other.

☐ Interfaces are clearly defined in terms of:

- signals
- data
- energy
- control

☐ The architecture shows information flow through the system.

☐ I explain how subsystem interactions enable the system to function.

☐ Interfaces are described conceptually, not electrically.

☐ Detailed communication protocols, circuits, or pin-level connections belong to Deliverable 3, not here.

☒ F. System Integration (High Level)

☐ I explain how all subsystems combine to form a working system.

☐ The system behaviour is described from input to output.

☐ The design shows how the system satisfies the user requirements defined in D1.

☐ Integration demonstrates systems-level reasoning.

☐ The design does not treat subsystems as isolated units.

☐ If the document describes subsystems but never explains how they work together → the system design is incomplete.

☒ G. What is deliberately NOT included

☐ No detailed circuit diagrams

☐ No PCB layouts

☐ No algorithms or code

☐ No component-level design

☐ No simulations or implementation results

☐ No bill of materials

☐ If any of these appear → you have moved into **Deliverable 3 or 4**.

☒ H. Engineering Reasoning Check

☐ The selected design is justified using engineering reasoning, not preference.

☐ Trade-offs are explicitly discussed.

☐ The design clearly addresses all key user requirements from D1.

☐ The architecture demonstrates systems engineering thinking.

☐ The document reads as a design argument, not a collection of diagrams.

☒ Engineering Reasoning Check

☐ I can explain each design option without reading from the document.

☐ I can defend why certain options were rejected.

☐ I can explain why the selected option is the best engineering choice.

☐ I can explain the purpose of each subsystem.

☐ I can describe how the entire system works from start to finish.

☐ If you cannot explain the design verbally, it will not be accepted as your own work.

 Final self-assessment

- ☐ My document shows engineering exploration before decision-making.
- ☐ My design selection is defensible and evidence-based.
- ☐ The system architecture is clear and logically structured.
- ☐ The design demonstrates engineering judgement and systems thinking.
- ☐ The document clearly shows the transition from problem (D1) to solution architecture (D2).

Submission rule

Deliverable 2 does not ask: “How will you build the system?” It asks: “Which engineering solution approach is most defensible and what does the system architecture look like?”